

Weekly training log, week 7

Week 7 to 8

Module	EEEM068, Applied Machine Learning
Project	Industrial waste classification on WaRP
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Focus	Swin Tiny on WaRP-C, four trials

1. Experimental settings

Dataset	WaRP-C, 8805 train, 1583 test
GPU	NVIDIA RTX 4000 Ada, 20 GB
Input size	224 x 224
Metric	Top 1 accuracy on EMA model, macro F1
Goal	Get Swin to work properly on WaRP-C.

2. Hyperparameters

Trial 1, heavy augmentation

Backbone	swin_tiny_patch4_window7_224, ImageNet pretrained
Batch size	32
Head LR	1e-4
Backbone LR	5e-5
Weight decay	0.05
Optimizer	AdamW with cosine LR
Loss	CE with class weights and label smoothing 0.1
EMA decay	0.999
Epochs	120 (25 head only, then 95 full)
Augmentation	crops [0.5, 1.0], colour jitter 0.4
MixUp	alpha 0.8, every batch

Trial 2, MixUp only in Phase 2

Change	Disabled MixUp during Phase 1, kept it for Phase 2.
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Trial 3, lighter MixUp plus CutMix

Change	MixUp alpha lowered to 0.4, added CutMix alpha 1.0 with probability 0.5.
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Trial 4, gentle pipeline, no MixUp

Epochs	80 (20 head only, then 60 full)
Augmentation	flips, rotations, crops [0.6, 1.0], colour jitter 0.2
MixUp	off

3. Results

Trial 1

Epoch	Loss	Train	Test (EMA)	Best
25	1.612	27.4%	27.45%	27.45%
55	1.342	68.7%	61.20%	62.18%
80	1.218	73.4%	62.74%	62.99%
120	1.181	75.1%	62.65%	62.99%

Best test accuracy: 62.99%. Macro F1: 0.599.

Trial 2

Epoch	Loss	Train	Test (EMA)	Best
25	1.601	28.0%	27.50%	27.50%
60	1.142	70.2%	64.18%	64.93%
90	1.089	72.8%	65.27%	66.41%
120	1.076	73.5%	65.84%	66.41%

Best test accuracy: 66.41%. About 3.4 points better than Trial 1.

Trial 3

Epoch	Loss	Train	Test (EMA)	Best
25	1.598	28.5%	27.91%	27.91%
60	1.118	71.9%	65.92%	66.93%
100	1.058	74.1%	68.10%	68.74%
120	1.048	74.4%	68.32%	68.74%

Best test accuracy: 68.74%. Still way below the CNN models.

Trial 4

Epoch	Loss	Train	Test (EMA)	Best
10	1.716	39.31%	18.89%	18.89%
20	1.675	42.79%	26.05%	26.05%
21	1.626	64.66%	37.14%	37.14%
30	0.885	92.58%	78.59%	78.59%
40	0.763	96.83%	80.92%	80.92%
49	0.715	98.29%	81.56%	81.56%
80	0.668	99.44%	81.17%	81.56%

Best test accuracy: 81.56%. Macro F1: 0.796. This is the version we keep.

4. Observations

The heavy MixUp recipe was wrecking Phase 1. The head never saw clean features so it never learned the classes properly.

Removing MixUp during the frozen phase already added 3 points (Trial 2). Lowering its strength helped a bit more (Trial 3) but it was still bad.

Trial 4 dropped MixUp entirely and used much lighter augmentation. Jumped to 81.56% on the same backbone. Big difference, only the augmentation changed.

Submitting Trial 4 as the Swin result for the report.